Question:

Jen enters a lottery by picking 4 distinct numbers from $S=\{1,2,3,\cdots,9,10\}.$ 4 numbers are randomly chosen from $S.$ She wins a prize if at least two of her numbers were 2 of the randomly chosen numbers, and wins the grand prize if all four of her numbers were the randomly chosen numbers. The probability of her winning the grand prize given that she won a prize is $\tfrac{m}{n}$ where m and n are relatively prime positive integers. Find $m+n$.

Reasoning + Answer:

<think>
1. We are given a problem involving a lottery where Jen picks 4 distinct numbers from a set $S = \{1, 2, 3, \cdots, 9, 10\}$. The numbers she picks and the numbers randomly chosen are all 4 distinct numbers from S.
2. Jen wins a prize if at least two of her numbers are the randomly chosen numbers.
3. Jen wins the grand prize if all four of her numbers are the randomly chosen numbers.
4. We need to find the probability that Jen wins the grand prize given that she won a prize.
5. Let's consider the event that Jen wins a prize, which means her numbers include at least 2 of the randomly chosen numbers.
6. For Jen to win the grand prize, all four of her numbers must be the randomly chosen numbers.
7. We need to calculate the conditional probability. The total probability that Jen wins a prize is the number of favorable outcomes divided by the total number of outcomes that result in winning a prize.
8. The number of ways to win a prize, which is having at least 2 of her numbers be the randomly chosen numbers, can be calculated by considering different cases:
9. Case 1: She has exactly 2 of her numbers be the randomly chosen numbers.
- The number of ways to choose 2 numbers out of the 4 chosen randomly (say $\{x, y, z\}$) to be Jen's numbers.
- We choose 2 out of the 4 chosen, and the remaining 2 can be chosen in the remaining choices which are 7 (after the 4 chosen) from 6 (the rest).

10. The number of ways to choose the 2 numbers out of the 4 chosen is $\binom{4}{2} = 6$, and the remaining 2 can be chosen in $7 \times 6 = 42$ ways.
11. Case 2: She has exactly 3 of her numbers be the randomly chosen numbers.
- This means all her 4 numbers are the chosen set. The number of ways to pick 4 out of 4 is 1.

12. We sum the cases to get the total number of favorable outcomes for winning a prize.

13. The total probability of winning a prize is the sum of the individual probabilities.
14. The probability of the grand prize, i.e., all 4 numbers being the same as the randomly chosen numbers, is the ratio of favorable outcomes to the total outcomes for winning a prize.
15. This probability can be simplified to a fraction $\frac{m}{n}$ where m and n are relatively prime.
16. Finally, we calculate $m+n$.
<answer>3</answer>